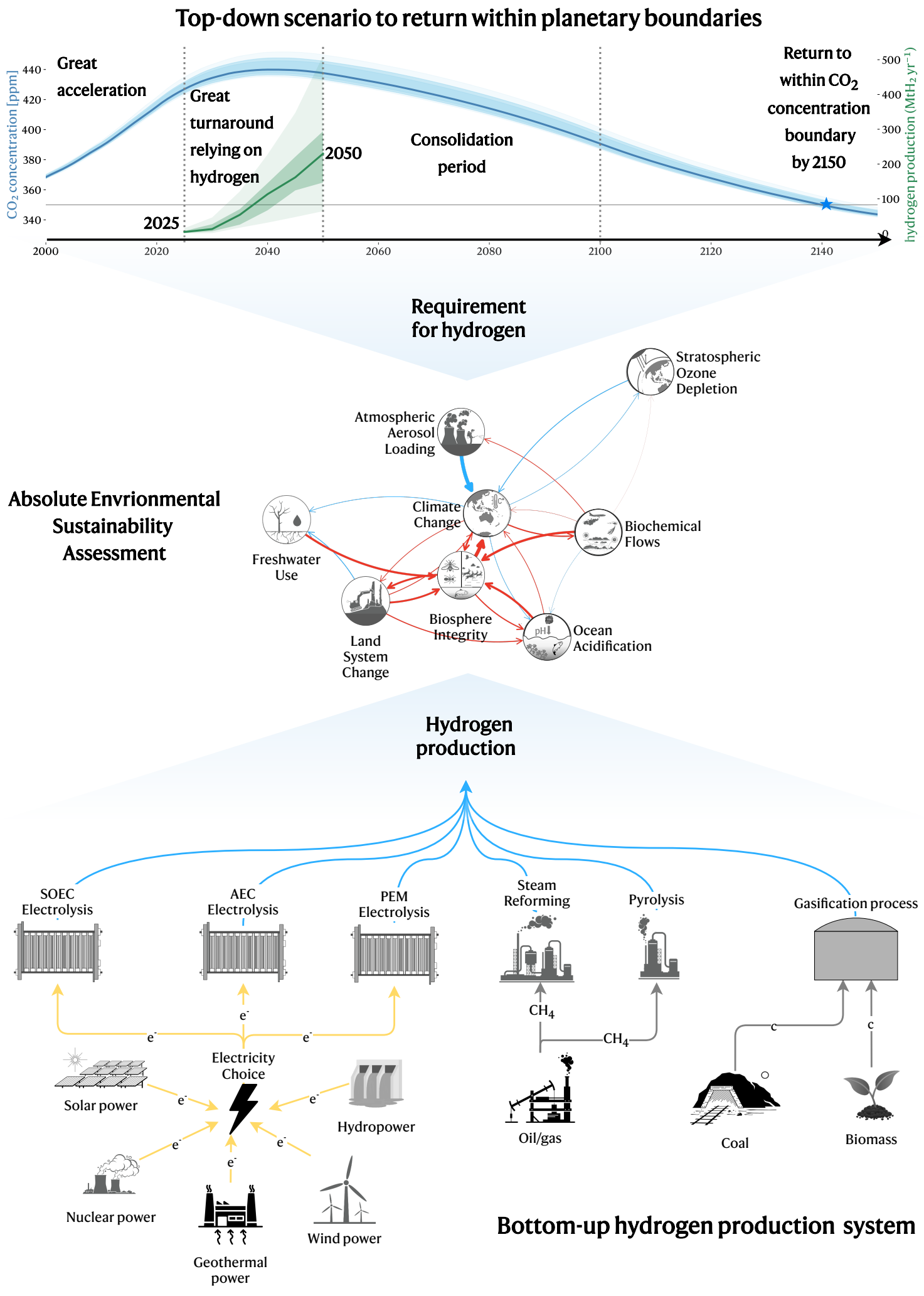
Top-down scenario to return within planetary boundaries
Great acceleration
Great turnaround relying on hydrogen
2025
2050
Consolidation period
Return to within CO_2 concentration boundary by 2150
CO_2 concentration [ppm]
hydrogen production ($MtH_2\,yr^{-1}$)
Requirement for hydrogen
Absolute Envrionmental Sustainability Assessment
Stratospheric Ozone Depletion
Atmospheric Aerosol Loading
Climate Change
Biochemical Flows
Freshwater Use
Biosphere Integrity
Land System Change
Ocean Acidification
Hydrogen production
SOEC Electrolysis
AEC Electrolysis
PEM Electrolysis
Steam Reforming
Pyrolysis
Gasification process
CH_4
Electricity Choice
Solar power
Hydropower
Oil/gas
Coal
Biomass
Nuclear power
Geothermal power
Wind power
Bottom-up hydrogen production system